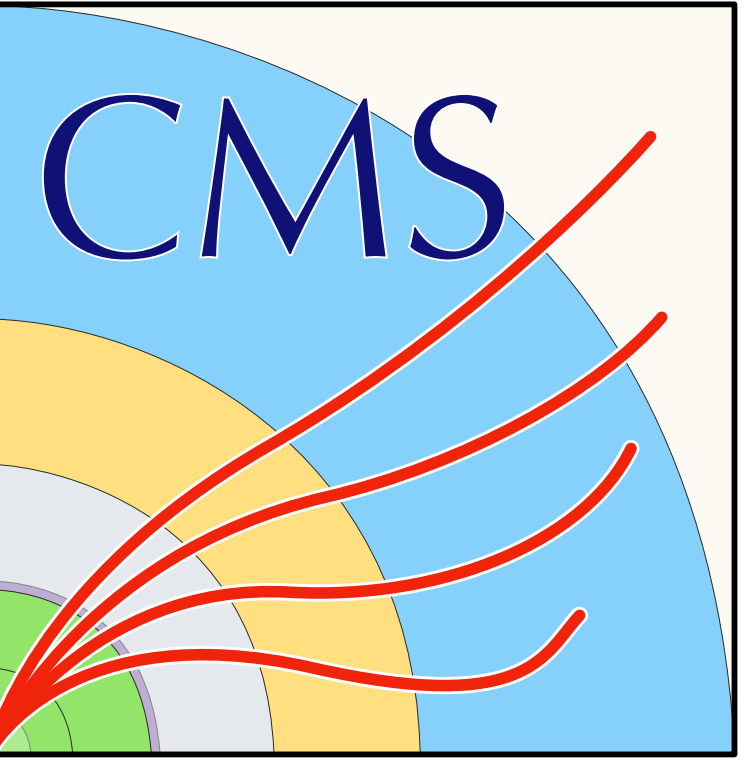




# Off-shell Higgs boson Production Measurements

Mohit Srivastav

Johns Hopkins University, CMS Joint 4-Lepton Study Team (CJLST)



## Objectives

Multiple measurements of off-shell Higgs boson production are presented:

- 1 The first test of Higgs substructure on CMS
- 2 The most precise indirect measurement of light-Yukawa couplings to date
- 3 Measurements of BSM phenomena in the EFT limit in gluon-fusion production
- 4 Constraints on the width with both heavy and light terms floated
- 5 The first measurement of  $\kappa_\lambda$  in the off-shell region
- 6 A combination of all the Run-2 off-shell width measurements to date

## The Off-shell Higgs boson

The off-shell region of the Higgs boson is present in VV decays above  $2m_V$ , and is characterized by a strong interference component with background. The hallmark of the off-shell region is the independence from the the width, as  $4 \text{ MeV} \ll 2m_V$ , from Eq. 1.

$$\frac{\sigma(i \rightarrow H \rightarrow f)}{ds} \propto \frac{1}{(s - M_H^2)^2 + M_H^2 \Gamma_H^2} \quad (1)$$

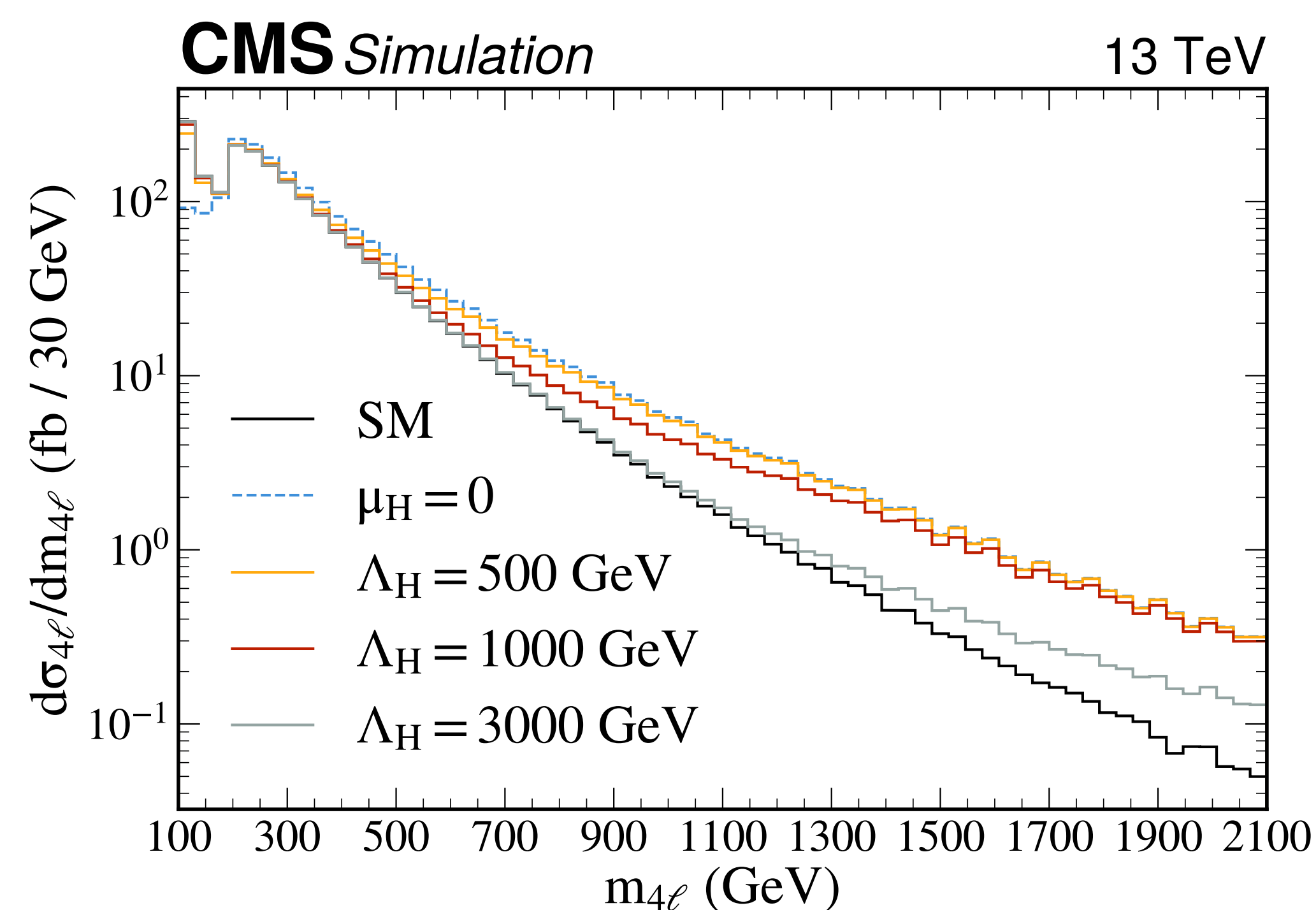


Figure 1: The EW evolution of the overall  $4\ell$  distribution with varying Higgs signal strengths

## Higgs Substructure

Figure 1 showcases the evolution given different values of  $\Lambda_H$  for the form factor described in Eq. 2, where  $\Lambda_H$  is the energy scale of the form factor, with the scan shown in Fig. 2

$$F(q^2) = \left[ \frac{1}{1 + |q^2|/\Lambda_H^2} \right]^2 \quad (2)$$

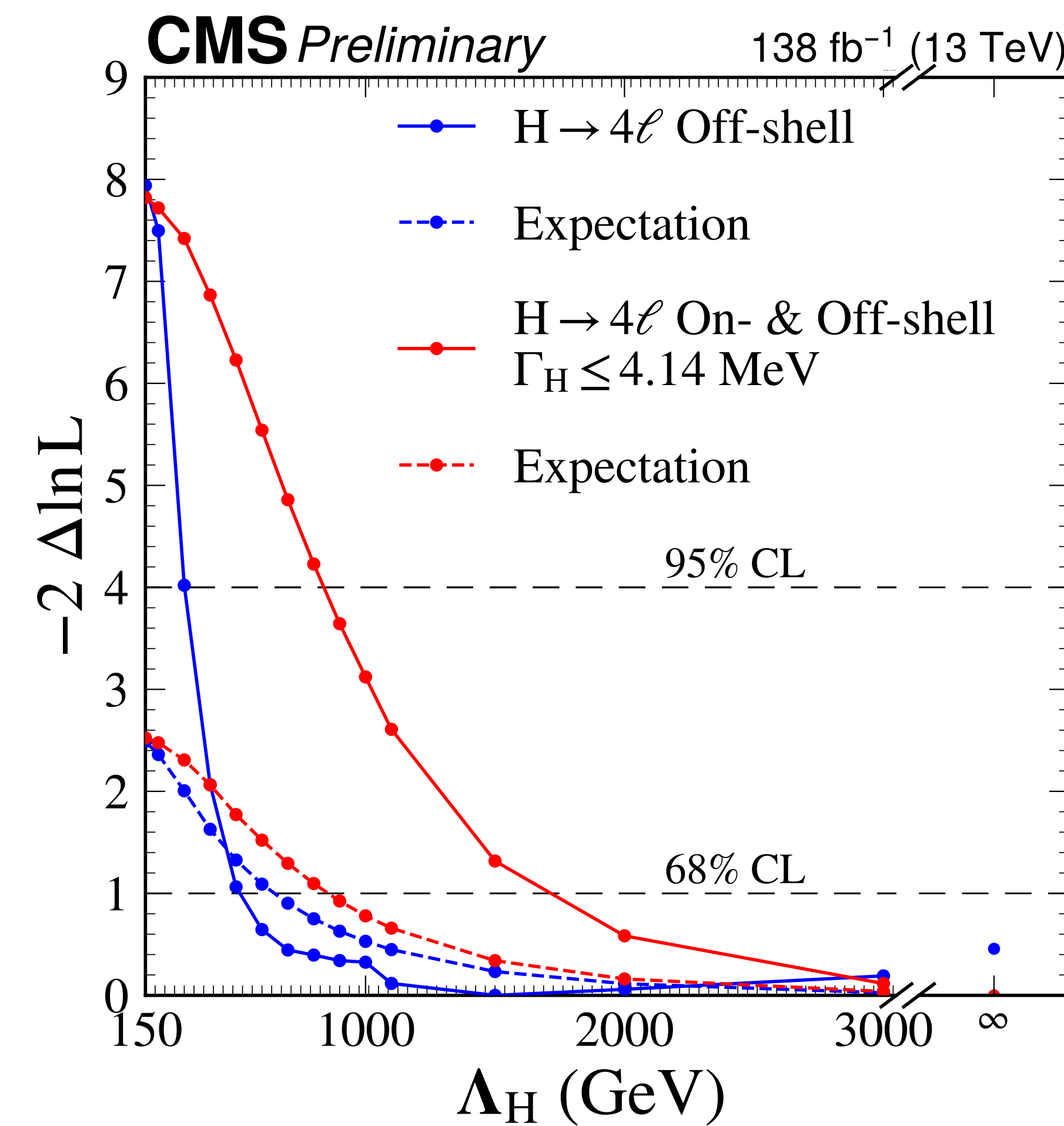


Figure 2: Scan for  $\Lambda_H$

## Mathematical Section

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$$E = mc^2 \quad (3)$$

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$$\cos^3 \theta = \frac{1}{4} \cos \theta + \frac{3}{4} \cos 3\theta \quad (4)$$

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## Methods

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## Results



Figure 3: Figure caption

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## Conclusion

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## Additional Information

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## References

- [1] J. M. Smith and A. B. Jones. *Book Title*. Publisher, 7th edition, 2012.
- [2] A. B. Jones and J. M. Smith. *Article Title*. *Journal title*, 13(52):123–456, March 2013.

## Acknowledgements

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## Contact Information

- Web: <http://www.university.edu/smithlab>
- Email: [john@smith.com](mailto:john@smith.com)
- Phone: +1 (000) 111 1111

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